

Test 5 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. Select **all** the numbers that are less than $-1\frac{1}{2}$.

- ☐ 0
- ☐ 2
- ☐ $\frac{3}{4}$
- ☐ $-2\frac{1}{2}$
- ☐ $-1\frac{3}{4}$
- ☐ $-\frac{1}{2}$

Select all that apply.

Show your work

calculator · multiselect

2. Select **all** the numbers with an absolute value greater than $3\frac{1}{2}$.

- ☐ -1
- ☐ 3
- ☐ $-6\frac{1}{2}$
- ☐ 7
- ☐ $-4\frac{1}{2}$
- ☐ $-2\frac{1}{2}$

Select all that apply.

Show your work

calculator · multiselect

3. Sofia's prepaid transit card shows a balance of 0 dollars.

What does the 0 mean in this situation?

- A. There is no money left on the card and no amount is owed.
- B. The card has \$28 of credit remaining.
- C. The card has been used 28 times.
- D. The card is worth more than it was yesterday.

Show your work

calculator · multiple choice

4. What is the solution to the equation $2x = 30$?
- A. 28
 - B. 60
 - C. 30
 - D. 15

Show your work

multiple choice

5. Match each triangle with its area, in square centimeters.

	14	27	30	33
base 7 centimeters, height 4 centimeters	☐	☐	☐	☐
base 9 centimeters, height 6 centimeters	☐	☐	☐	☐
base 6 centimeters, height 10 centimeters	☐	☐	☐	☐

Show your work

matching table

6. What is the value of $-7 + (25)$?
- A. -18
 - B. 32
 - C. 18
 - D. -32

Show your work

calculator · multiple choice

7. A rectangular garden is drawn on a coordinate plane with vertices $(-4, -5)$, $(1, -5)$, $(1, 1)$ and $(-4, 1)$. Each unit is 1 meter.

What is the perimeter of the garden, in meters? Enter your answer.

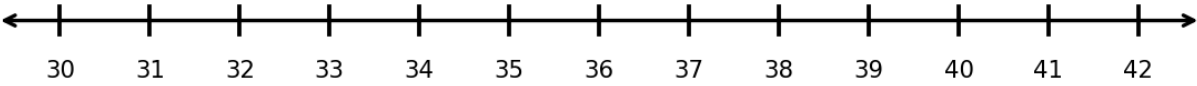
Answer: _____

Show your work

equation editor

8. To ride the roller coaster at Kendall Indian Hammocks, a rider must be at least 36 inches tall.

Plot the boundary value of the solution on the number line.



Mark your answer on the diagram above.

Show your work

graphing (grid)

9. What is the product of $\frac{2}{3}$ and $\frac{1}{2}$?

- A. $\frac{1}{3}$
- B. $1\frac{1}{6}$
- C. $1\frac{1}{3}$
- D. $\frac{3}{5}$

Show your work

calculator · multiple choice

10. Match each expression with an equivalent expression.

	$2x$	$2x + 4$	$4x$	$9x$
$2(x + 2)$	☐	☐	☐	☐
$2x + 7x$	☐	☐	☐	☐
$2x + 2 - 2$	☐	☐	☐	☐

Show your work

matching table

11. What is the product of 45.9 and 0.45?

- A. 2065.5
- B. 20.655
- C. 2.0655
- D. 206.55

Show your work

calculator · multiple choice

12. What is the solution to the equation $x + 8 = 41$?

- A. 8
- B. 49
- C. 41
- D. 33

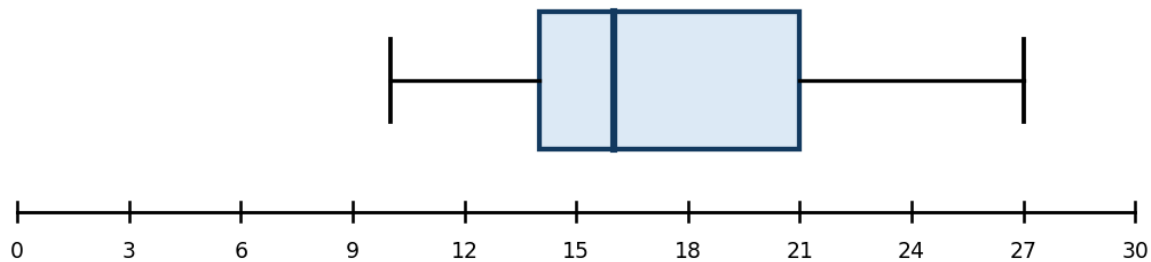
Show your work

multiple choice

13. The box plot shows a set of data.

What is the median of the data?

Daily high temperature (°F)



- A. 21
- B. 17
- C. 16
- D. 14

Show your work

multiple choice

14. The table shows equivalent ratios of red beads to blue beads.

What number belongs in the box marked ?

Red	Blue
3	4
6	8
15	?

Equivalent ratios

- A. 9
- B. 25
- C. 15
- D. 20

Show your work

multiple choice

15. Camila buys 8 rolls of tape at \$4.11 each and pays a \$5.95 delivery fee.

What is the total cost?

- A. \$80.48
- B. \$32.88
- C. \$38.83
- D. \$51.71

Show your work

calculator · multiple choice

16. Which question is a statistical question?

- A. What time does the first bell ring at my school?
- B. How old is my teacher?
- C. How many books did I read last month?
- D. What are the heights, in inches, of the students in the band?

Show your work

multiple choice

17. Which expression shows the prime factorization of 126?

- A. 2^7
- B. $2^2 \times 3^2 \times 7$
- C. $2 \times 3^2 \times 7$
- D. $3^2 \times 7$

Show your work

calculator · multiple choice

18. What is the value of 6^4 ?

- A. 30
- B. 24
- C. 216
- D. 1296

Show your work

calculator · multiple choice

19. Which percent is equivalent to $\frac{3}{20}$?

- A. 15%
- B. 666.6667%
- C. 25%
- D. 1.5%

Show your work

calculator · multiple choice

20. What value of x makes the equation $x + 5.98 = 9.63$ true?

- A. 15.61
- B. 3.65
- C. 5.98
- D. 9.63

Show your work

multiple choice

21. The line plot shows books read last month reported by a group of students.

How many students reported 4 or more?



- A. 8
- B. 11
- C. 9
- D. 13

Show your work

multiple choice

22. The table shows the number of minutes six students spent on the bus.

What is the mean number of minutes?

Minutes	#1	#2	#3	#4	#5	#6
Minutes	17	21	22	24	27	27

- A. 24
- B. 10
- C. 138
- D. 23

Show your work

multiple choice

23. A store sells 8 pounds of avocados for \$48.00.

What is the unit price, in dollars per pound?

- A. \$1.33
- B. \$56.00
- C. \$48.00
- D. \$6.00

Show your work

multiple choice

24. A banner is 36 feet long.

How many inches long is the banner? (1 foot = 12 inches)

- A. 432 inches
- B. 48 inches
- C. 3 inches
- D. 108 inches

Show your work

multiple choice

25. Match each written description with the expression it represents.

	$(x + 11)/2$	$11x$	$11x + 11$	$x - 11$
a number x decreased by 11	■	■	■	■
the sum of a number x and 11, divided by 2	■	■	■	■
the product of 11 and a number x	■	■	■	■

Show your work

matching table

26. There are 120 students in the sixth grade at Palmetto Middle. 40% of them ride the bus.

How many students ride the bus?

- A. 3
- B. 48
- C. 80
- D. 480

Show your work

multiple choice

27. A change of more than 10 degrees from 0 sets off an alarm.

Select **all** the readings that set off the alarm.

- ☐ -12
- ☐ 6
- ☐ 11
- ☐ -10
- ☐ -7
- ☐ 1

Select all that apply.

Show your work

calculator · multiselect

28. What is the value of $(8)(8)$?

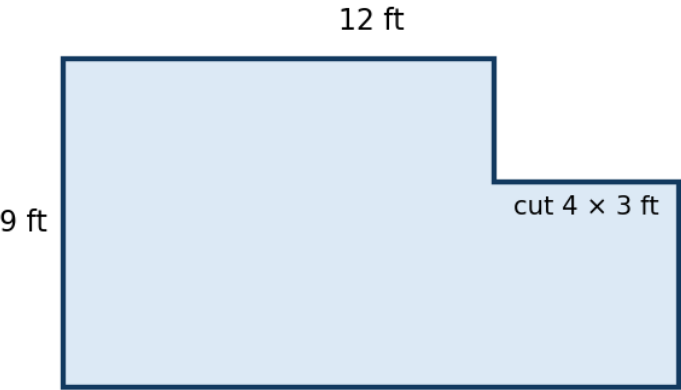
- A. 16
- B. -64
- C. 64
- D. 68

Show your work

calculator · multiple choice

29. Use the drop-down menus to find the area of the figure, in square feet.

The area of the whole rectangle is square feet, and the area of the cut-out corner is square feet.



Blank (1) choices: 21 | 42 | 96 | 108

Blank (2) choices: 7 | 12 | 14 | 36

Show your work

editing task choice

30. A data set is shown.

15, 20, 21, 21, 23

The value 28 is added to the data set. What is the mean of the new data set?

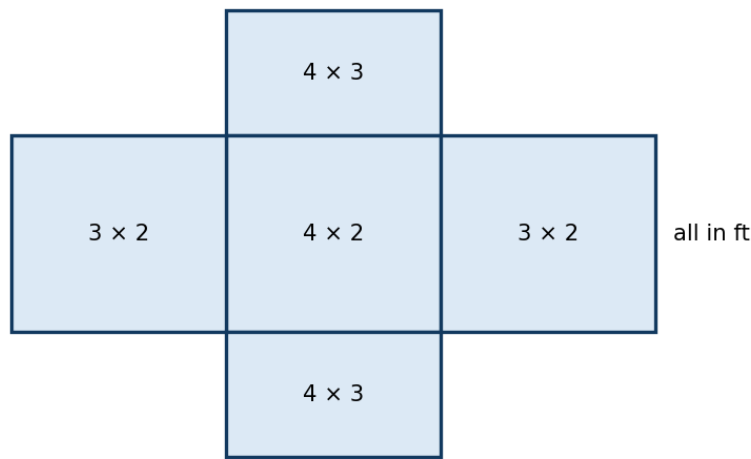
- A. $25 \frac{3}{5}$
- B. $21 \frac{1}{3}$
- C. 20
- D. 28

Show your work

multiple choice

31. The net of a right rectangular prism is shown.

What is the surface area of the prism, in square feet?



- A. 24
- B. 26
- C. 58
- D. 52

Show your work

multiple choice

32. Match each point with the quadrant it is located in.

	Quadrant I	Quadrant II	Quadrant III	Quadrant IV
D (5, -2)	■	■	■	■
A (3, 4)	■	■	■	■
C (-4, -3)	■	■	■	■

Show your work

matching table

33. What is the greatest common factor of 18 and 30?

- A. 12
- B. 18
- C. 90
- D. 6

Show your work

calculator · multiple choice

34. Each expression is evaluated for $x = 8$. Match each expression with its value.

	-4	48	52	384
$6x + 4$	☐	☐	☐	☐
$6x^2$	☐	☐	☐	☐
$4 - x$	☐	☐	☐	☐

Show your work

matching table

35. Select **all** the expressions with a value of 2.4.

- ☐ $24 \div 10$
- ☐ 2.5×1
- ☐ 1.2×2
- ☐ 0.4×6
- ☐ $2.4 \div 0.1$
- ☐ 2.4×10

Select all that apply.

Show your work

calculator · multiselect

36. A sixth-grade class at Glades Middle has 36 students who ride the bus and 28 students who walk.

What is the ratio of students who ride the bus to students who walk, in simplest form?

- A. 16:7
- B. 9:7
- C. 7:9
- D. 9:16

Show your work

multiple choice

37. Rewrite $12 + 30$ as the greatest common factor of the two numbers multiplied by the sum of two whole numbers.

Which expression is correct?

- A. $6(7)$
- B. $6(2 + 5)$
- C. $2(6 + 5)$
- D. $6(2 + 6)$

Show your work

calculator · multiple choice

38. A data set is shown.

2, 7, 11, 17, 20, 23, 26, 34, 37

Which list gives the five-number summary (minimum, lower quartile, median, upper quartile, maximum) of the data?

- A. 2, 20, 9, 30, 37
- B. 2, 9, 17, 30, 37
- C. 2, 7, 20, 34, 37
- D. 2, 9, 20, 30, 37

Show your work

multiple choice

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume